

DENDRITRON RECURRENT ENGRAM

Sparse Memory Carries Knowledge.
Two Blocks Carry Thought.

A CPU/ARM-first architecture joining frozen phrase and definition memory, JTD-aligned LNgram routing, signed HarMax geometry, and composed LoRA skills.

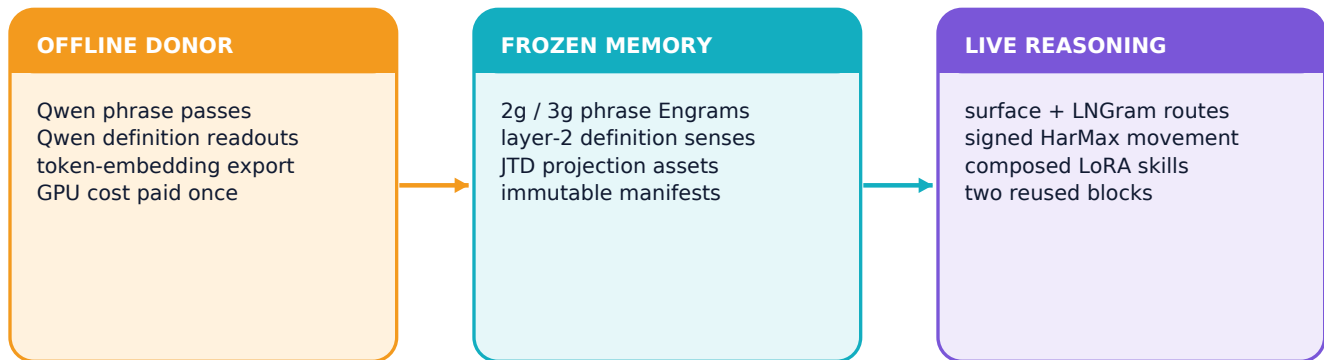
PHRASE ENGRAMS	DEFINITION BANK	LIVE CORE
1,000,000 exact rows layer08[2048] BF16 layer24[2048] BF16 separate 2g / 3g banks	1,532,746 sense rows layer02[2048] BF16 31 immutable shards canonical JTD frame	2 stored blocks effective depth 2R composed LoRA skills bounded experts

Primary intent: pay for the donor once.
Retrieve knowledge sparsely during live CPU/ARM execution.

01 | INTENT

Freeze knowledge. Reuse reasoning.

A FROZEN DONOR-BUILT MEMORY PLANE FEEDS A DISTINCT EVOLVING CPU/ARM STATE



Dendritron separates stable semantic memory from changing thought. Frozen phrase vectors and definition-sense landmarks are addressed sparsely. The evolving 2,048D live state carries the current problem, intermediate reasoning, and task context through repeated visits to two physical blocks.

MEMORY PLANE

Exact suffix lookup, Hash-Engram miss coverage, LNGram definition addressing, and immutable donor rows. Retrieved values enter the live state through calibrated gated residuals.

COMPUTE PLANE

Two DeepLoop blocks apply HarMax contraction followed by composed skill LoRAs and sparse expert computation. Stored depth is 2; effective thought depth is 2R.

RESEARCH BASIS AND DENDRITRON EXTENSION

Conditional Memory and Memory Grafting motivate static lookup as a separate sparsity axis. DeepLoop supplies visit-aware recurrent scaling. Dendritron joins these with a frozen layer-2 semantic frame, JTD alignment, LNGram latent addresses, signed HarMax evidence, and composed shared/private LoRA skills.

02 | BUILT ASSETS

Built memory and alignment assets

PUNCTUATION-V2 PHRASE BANKS, DEFINITION LANDMARKS, TOKEN SEEDS, AND REAL CPU JTD FITS

TRIGRAM ENGRAM

500,000 exact 3-word rows
layer08[2048] + layer24[2048]
independent row and shard space

BIGRAM ENGRAM

500,000 exact 2-word rows
layer08[2048] + layer24[2048]
independent row and shard space

Asset	Measured build evidence	State
Punctuation-v2 inventory	421,332 bigrams and 388,443 trigrams retained; 78,668 and 111,557 replaced	COMPLETE
Corrected phrase bank	809,775 paired rows copied + 190,225 freshly extracted = 1,000,000 rows	COMPLETE
Qwen symbol seeds	248,077 x 2,048 BF16 input-embedding rows	COMPLETE
Definition landmarks	1,532,746 x 2,048 BF16 layer-2 sense rows; 31 shards; about 5.8 GiB	REPORTED
JTD anchor triples	32,768 bigrams + 32,768 trigrams; layer-2 reference with layer-8/24 sources	COMPLETE
Real CPU JTD fit	J8 loss 21.849 to 0.372; J24 loss 37.923 to 0.844	COMPLETE

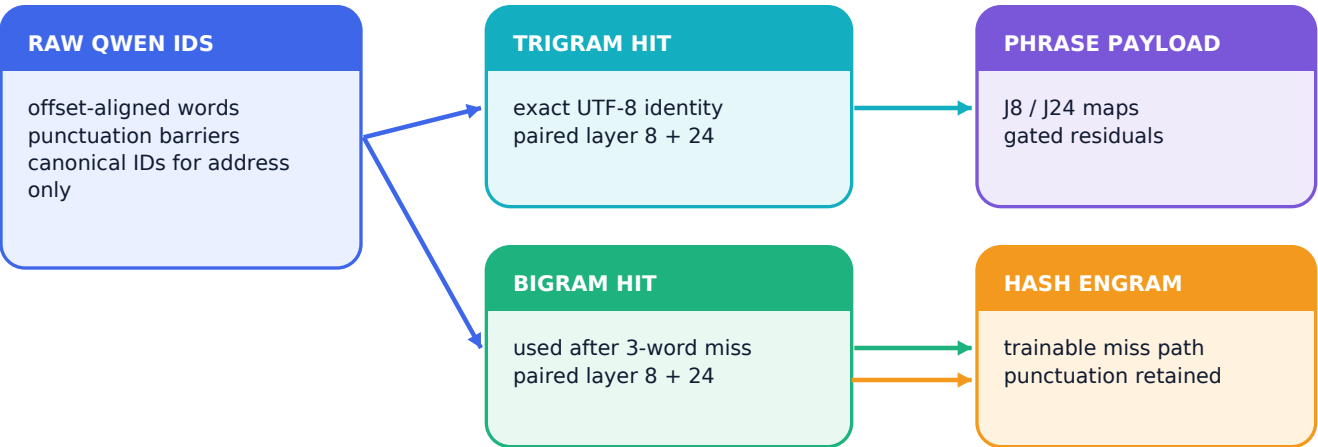
ASSET LIFECYCLE

Canonical commits reference frozen assets by manifest identity, revision, shape, dtype, and cryptographic hash. New Engram, LoRA, router, expert, and provenance deltas remain compact and versioned.

03 | SURFACE MEMORY

Exact phrases and constituent senses

LONGEST EXACT 3-TO-2 PHRASE LOOKUP COEXISTS WITH PER-WORD DEFINITION SEEDING



CONSTITUENT SENSE SEEDING

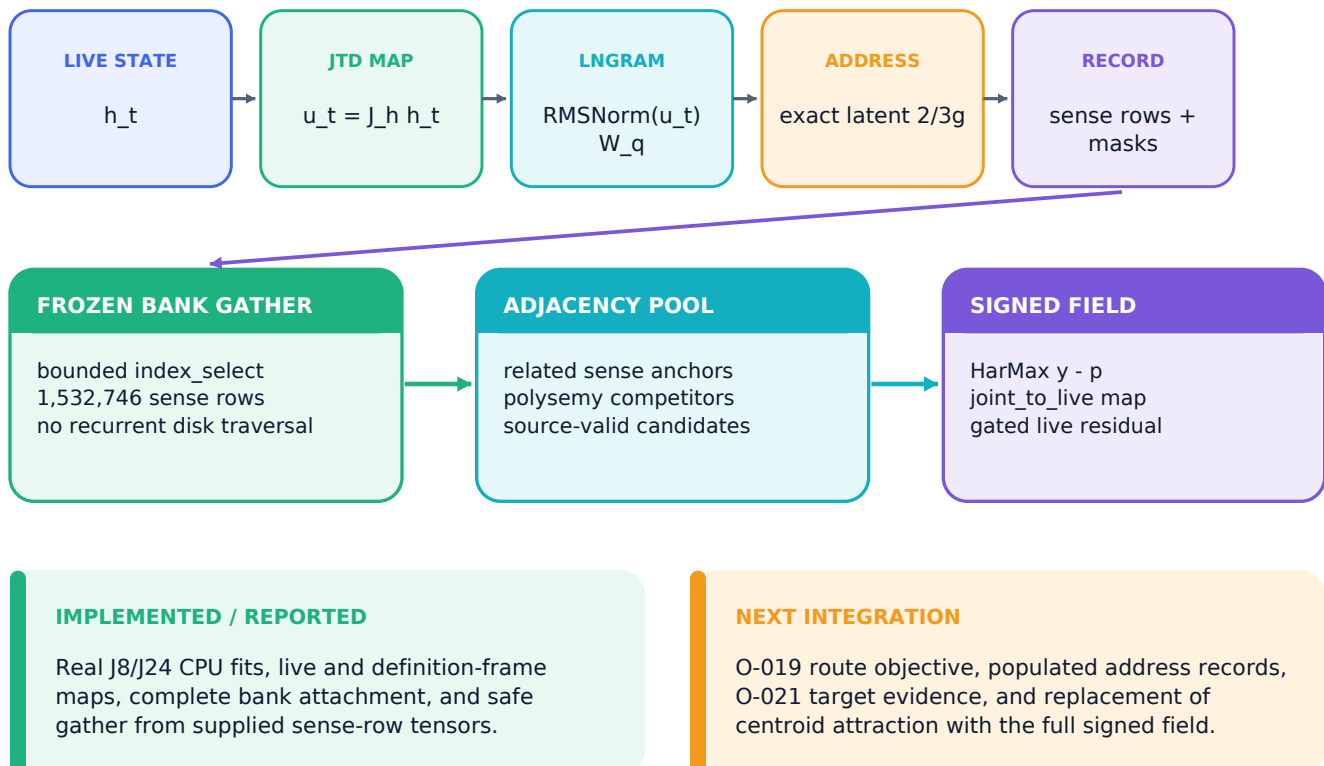
Every complete input or decoded-reasoning word emits its dictionary sense IDs into the bounded definition pool, even when a longer exact phrase supplies the primary phrase-memory row. Sense IDs are deduplicated while source, span, and word-order provenance remain attached.

Layer	Address	Payload	Role
Phrase	Exact 3-word, else exact 2-word	Frozen layer-8 and layer-24 vectors	Narrow contextual memory
Constituent	Every complete word	All retained definition-sense handles	Semantic neighborhood seeds
Miss	Canonical local Qwen-ID suffix	Trainable Hash-Engram rows	Coverage outside frozen phrase bank

04 | DEFINITION MEMORY

JTD-aligned LNGram definition memory

CONTINUOUS ALIGNMENT, DISCRETE SPARSE SELECTION, THEN CONTINUOUS EVIDENCE MOVEMENT



The latent address integer and dictionary row integers remain separate namespaces. JTD places the evolving live query in the frozen layer-2 definition frame before LNgram forms its address. The retrieved vectors already occupy that frame, so the bridge performs selection and gathering rather than a second semantic map.

RESEARCH BASIS AND DENDRITRON EXTENSION

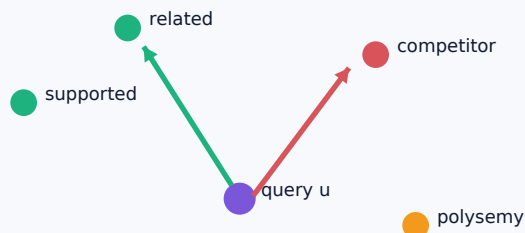
LNgram contributes tokenizer-independent discrete routing from continuous hidden states. Locality Preserving Joint Transfer contributes source-specific continuous alignment. Dendritron binds them through bounded definition-sense handles and keeps phrase Engrams and Hash-Engram as independent memory branches.

05 | GEOMETRIC CONTROL

Evidence moves the live state

HARMAX TARGET MASS MINUS PROXIMITY MASS CREATES SIGNED ATTRACTION AND REPULSION

JOINT-SPACE CONTRACTION POOL



SIGNED FIELD

$$d_q^2 = \|u - c_q\|_2^2 + \text{eps}^2$$

$$p_q = d_q^{-1} / \sum_r d_r^{-1}$$

$$\Delta u = n \sum_q (y_q - p_q)(c_q - u) / d_q^2$$

$$h \leftarrow h + \text{gate} * \text{joint_to_live}(\Delta u)$$

y > p

Underrepresented supported evidence attracts the live query.

y < p

Unsupported or excess proximity produces repulsion.

y = 0

Competitors remain in the denominator and supply contrast.

CURRENT BOUNDARY

The recurrent HarMax core already implements target and distance mass, attraction, repulsion, and the harmonic residual. The definition branch currently performs inverse-distance centroid attraction. O-021 defines the target evidence y needed to connect this complete signed operator to the gathered definition pool.

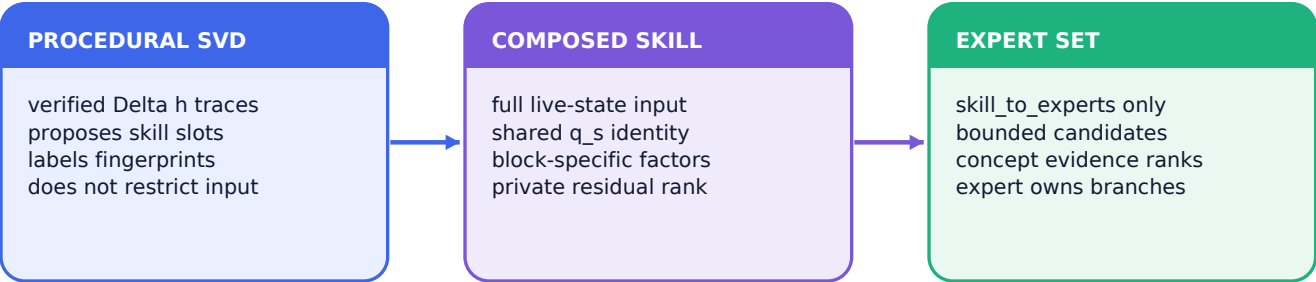
06 | PROCEDURAL ROUTING

Composed skills, bounded experts

EACH SKILL COMBINES A BLOCK-SPECIFIC SHARED TERM WITH ITS OWN PRIVATE RESIDUAL TERM

COMPLETE SKILL LORA FOR SKILL *s* AND BLOCK *b*

$$\Delta W_{s,b} = B_b^{sh} \text{diag}(q_s) A_b^{sh} + B_{s,b}^{priv} A_{s,b}^{priv}$$



Element	Contract
Shared term	Block-specific A/B factors; dense <i>q_s</i> spans the full shared basis and preserves skill identity across blocks.
Private term	Every populated skill owns block-specific private factors; rank 4 is provisional and selected from residual energy.
Capacity	Twenty-one current slots are placeholder capacity. Populated and active masks control actual use.
Router boundary	LNGram addresses definition memory only. Skill adjacency selects experts independently.

CURRENT TREE: 176 TESTS REPORTED

07 | CONTINUAL LEARNING

Frozen runtime, verified offline updates

SUCCESSFUL TRAJECTORIES BECOME PROVENANCE-BEARING EXPERIENCE AND CAREFULLY VALIDATED PARAMETER DELTAS



LIVE EXECUTION

Core weights, shared basis, skill coefficients, private factors, experts, and router anchors are frozen during inference and task execution.

OFFLINE WORKING BUFFER

Only verified success authorizes a private-skill update. Replay, regression, provenance, and version checks precede commit.

CANONICAL ASSET-REFERENCED CHECKPOINT

Stores compact changing state

- new or updated Engram records
- composed shared/private LoRA factors and q coefficients
- router, expert, and branch registries
- success evidence, verification, and provenance

References immutable assets once

- token-embedding matrix
- definition bank
- layer-8/layer-24 phrase banks
- fitted JTD projections
- identity, revision, shape, dtype, and hash

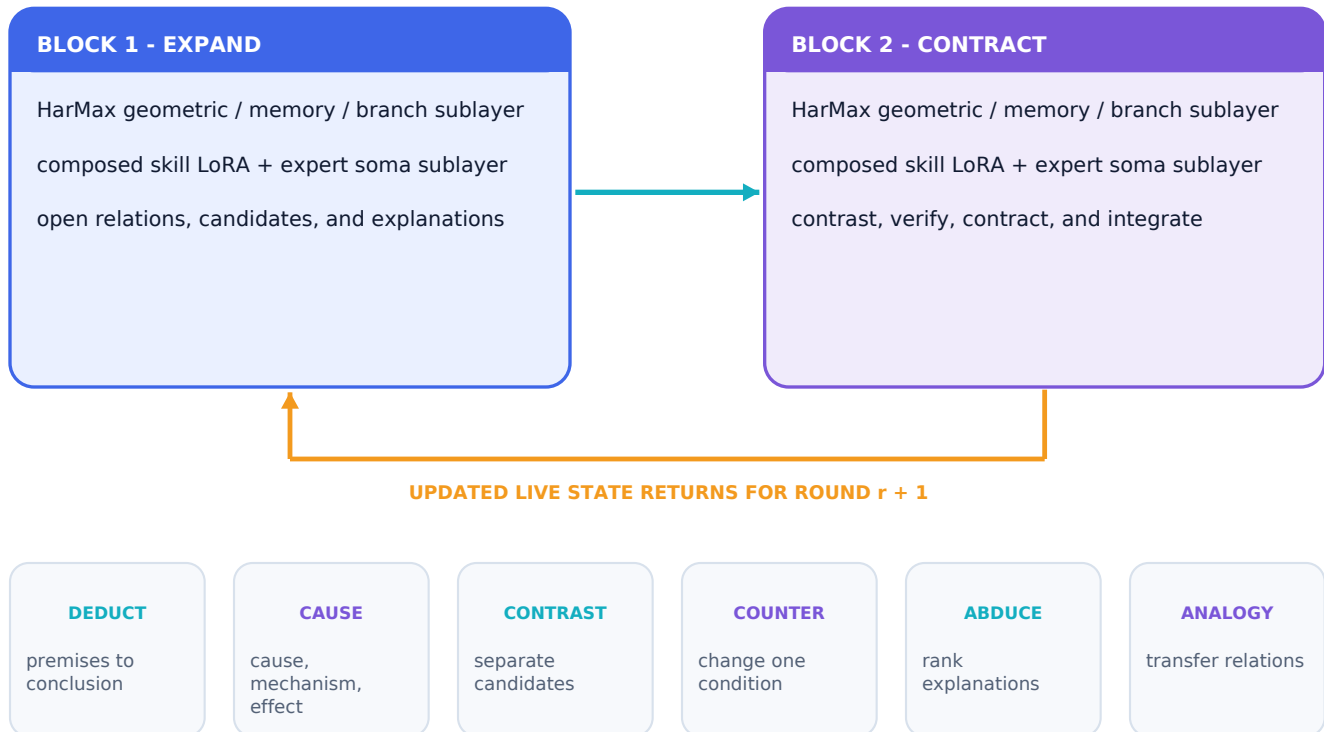
CONTENT AND PROCEDURE STAY SEPARATE

Personal facts and successful outcome traces belong in experience Engrams. Shared and private LoRA factors carry procedural skill. Frozen lexical and phrase assets retain byte identity across runtime and offline fitting.

08 | REASONING ENGINE

Two blocks carry recurrent thought

EACH ROUND EXPANDS CANDIDATES, THEN CONTRACTS THEM AGAINST MEMORY, EVIDENCE, AND TYPED BRANCH STATE



DEEPLoop STABILITY CONTRACT

Stored depth = 2. Effective depth = $2R$. Every residual-sublayer visit uses Post-RMSNorm with $\alpha = \sqrt{2N}$; designated residual matrices use initialization gain $\beta = 1 / \sqrt{8N}$. Visit alignment κ_R remains a measured stability diagnostic.

IMPLEMENTATION BOUNDARY

The two-block CPU reference, recurrent unroll, HarMax gradients, and adaptive relative-change stopping are verified. The six typed branch contracts are specified; complete live executors and task-level measurements remain engineering milestones.

09 | EXECUTION CONTRACT

CPU/ARM execution and the 25/75 law

THE LIVE PATH FETCHES BOUNDED MEMORY AND EXECUTES BOUNDED CONDITIONAL COMPUTATION

FIXED SPARSE-CAPACITY LEDGER

25% MEMORY

75% CONDITIONAL COMPUTE

OFFLINE GPU

Qwen phrase passes, definition readouts, and token-embedding export.

LIVE CPU / ARM

JTD, LNGram, sparse gathers, HarMax, LoRAs, experts, blocks, and decoding.

BOUNDED MEMORY

Attach frozen banks once; gather selected rows per visit; keep SQLite/shards outside the unroll.

SHARED METRIC

Ordinary Euclidean distance controls HarMax, concept scoring, route evaluation, and vocabulary ranking.

Runtime choice	Live effect
Exact address and bounded gather	Memory-bank size does not expand the active candidate pool.
Two reused blocks	Reasoning depth grows through visits rather than stored layers.
Low-rank skills + sparse experts	Only the routed procedural neighborhood executes.
Softmax-free operators	Routing, memory fusion, expert selection, and tied Euclidean decoding share a CPU-friendly contract.

10 | HONEST BOUNDARY

Built boundary and next integration

COMPLETED ASSETS AND VERIFIED MODULES STAY DISTINCT FROM THE REMAINING DEFINITION-ROUTE ASSEMBLY

Punctuation-v2 phrase inventories and 1M paired donor rows

COMPLETE

Token seeds, JTD anchor export, and real CPU J8/J24 fits

COMPLETE

1,532,746-row definition bank and supplied-sense-row attachment

REPORTED

Two-block HarMax core and tied Euclidean output

CPU VERIFIED

Composed LoRA / expert / provenance current tree

176 TESTS REPORTED

LNGram address to populated bounded sense-row records

NEXT: O-019

Definition-pool target evidence and full signed field

NEXT: O-021

Typed branch executors, language training, and ARM benchmarks

MILESTONE

IMMEDIATE BUILD ORDER

1. Select and verify O-019 routing supervision. 2. Populate bounded address records and measure route stability. 3. Lock O-021 evidence construction and connect the signed definition field. 4. Prove end-to-end constituent-sense payload materialization. 5. Train, benchmark, and optimize the typed branch system on ARM.

PRIMARY REFERENCES

[1] Cheng et al. Conditional Memory via Scalable Lookup. arXiv:2601.07372. [2] Cheng et al. Memory Grafting. arXiv:2605.20948. [3] Zheng et al. Lngam. arXiv:2605.24869. [4] Li et al. Locality Preserving Joint Transfer. arXiv:1906.07441. [5] Kaushik et al. Shared LoRA Subspaces. arXiv:2602.06043. [6] Li et al. DeepLoop. arXiv:2607.13491. [7] Lin. Collision-Free Hot-Tier Extension. arXiv:2601.16531. [8] Li. User as Engram. arXiv:2606.19172.

CREDITS AND CONTRIBUTION BOUNDARY

Ryan Carson's contribution is the Dendritron Recurrent Engram synthesis, subsequent extensions, integration contracts, project ledger, and validation coordination. The initial Dendritron starter and ARM package are pre-existing foundations and are excluded from that attribution. The cited papers remain work of their authors.